

CSCI 160 Homework 3 — Answer Key

1. Truth table and K-map for $F(x, y, z) = x'z + yz + xy'$.

Truth table. Evaluate each term and OR them:

x	y	z	$x'z$	yz	xy'	F
0	0	0	0	0	0	0
0	0	1	1	0	0	1
0	1	0	0	0	0	0
0	1	1	1	1	0	1
1	0	0	0	0	1	1
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	1	0	1	0	1

The minterms where $F = 1$ are $\sum m(1, 3, 4, 5, 7)$.

K-map and grouping.

$x \backslash yz$	00	01	11	10
0	0	1	1	0
1	1	1	1	0

xy' z

$F = z + xy'$

2. $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 11, 14)$.

Prime implicants (all four turn out to be *essential*):

Loop	Cells (minterms)	Term
blue	0, 2, 8, 10	$B'D'$
red	2, 3, 6, 7	$A'C$
green	2, 3, 10, 11	$B'C$
purple	2, 6, 10, 14	CD'

The four maps below show each loop on the same grid:

$AB \backslash CD$	00	01	11	10	
00	1	0	1	1	$A'C$
01	0	0	1	1	
11	0	0	0	1	$B'D'$
10	1	0	1	1	
					CD'

Essential check. $m_0 \rightarrow B'D'$, $m_7 \rightarrow A'C$, $m_{11} \rightarrow B'C$, $m_{14} \rightarrow CD'$: each is covered by a single loop, so all four are essential.

$$F = B'D' + A'C + B'C + CD'$$

3. $F(A, B, C, D) = \prod M(1, 3, 5, 7, 9, 11, 13, 15)$, as a product of sums.

For a POS we group the 0's (the listed maxterms). Every maxterm index is odd, i.e. has $D = 1$, so all eight 0's form one group.

$AB \backslash CD$	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

zeros: $D = 1$

The grouped 0's give $F' = D$, hence $F = D'$.

$$F = D'$$

4. $G(w, x, y, z) = \sum m(0, 1, 4, 5, 12, 13)$ in (a) SOP and (b) POS. Here the variable order is w, x, y, z , so a minterm index $= 8w + 4x + 2y + z$; the map rows are wx and columns yz .

(a) Sum of products — group the 1's.

$AB \backslash CD$	00	01	11	10
00	1	1	0	0
$w'y'$ 01	1	1	0	0
xy' 11	1	1	0	0
10	0	0	0	0

$$G = w'y' + xy'$$

(b) **Product of sums** — **group the 0's.** The 0's split into two groups: $y = 1$ (columns $yz = 11, 10$: minterms 2, 3, 6, 7, 10, 11, 14, 15), and wx' (row $wx = 10$: minterms 8, 9, 11, 10). These give the factors of G' .

$AB \backslash CD$	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
wx' 10	0	0	0	0

y

So $G' = y + wx'$. Complementing (DeMorgan):

$$G = (y + wx')' = y'(w' + x).$$

$$G = (w' + x) y'$$

5. $F(w, x, y, z) = \sum m(3, 4, 5, 6, 7)$ with $d(w, x, y, z) = \sum(10, 11, 12, 13, 14, 15)$. Minimal SOP. Again index = $8w + 4x + 2y + z$; don't-cares are marked $\times$ and may be used as 1 whenever they enlarge a group.

$AB \backslash CD$	00	01	11	10
00	0	0	1	0
01	1	1	1	1
x 11	$\times$	$\times$	$\times$	$\times$
10	0	0	$\times$	$\times$

yz

The x loop (middle two rows) uses don't-cares 12, 13, 14, 15; the yz loop (column $yz = 11$) uses don't-cares 11, 15. Together they cover 3, 4, 5, 6, 7.

$$F = x + yz$$